

Sub Code: BCAT007

ROLL NO. 24162900007

EVEN SEMESTER EXAMINATION, 2024 – 25

First Year (II Sem) B.C.A.

Computer Organization and Architecture

Duration: 3:00 hrs

Max Marks: 100

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i)What is floating-point representation? How is arithmetic operations performed on floating-point numbers? (ii)Represent the signed decimal number -25 in: 1) Signed magnitude 2) 1's complement 3) 2's complement (using 8 bits) b) Describe the Von Neumann architecture with a neat diagram. Explain how it works. c) Discuss the concept of multiprocessors and multicomputer systems. What are the differences?
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i)Explain the instruction cycle in detail with a diagram showing Fetch, Decode, Execute, and Interrupt stages. (ii)Discuss the differences between RISC and CISC architectures with advantages and disadvantages of each. b) Describe the different types of addressing modes with examples. c) What is stack organization in CPU design? Explain how subroutine calls and returns are handled using stack.
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i)Draw the logic diagram of an Arithmetic Logic Shift Unit (ALU) and explain how it performs arithmetic, logic, and shift operations using a mode control. (ii)Explain with block diagram how address sequencing is handled in a control memory unit during execution of microinstructions. Include subroutine calls, conditional branching, and return mechanisms. b) Design a bus system for four 4-bit registers using multiplexers. Explain how data is transferred between registers using control lines. c) Differentiate between hardwired control and microprogrammed control. Under what conditions would you choose one over the other? Justify with examples.
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i)Explain the memory hierarchy in computer systems. How does it help in improving system performance? Illustrate with an example involving cache, RAM, and secondary storage. (ii) Describe the organization of cache memory. b) What is virtual memory? Explain how paging works in virtual memory systems with a diagram. What are the advantages and drawbacks of using paging? c) A system has a cache hit ratio of 90%, cache access time of 10 ns, and main memory access time of 100 ns. • i) Calculate the effective memory access time.

	• ii) If the hit ratio drops to 80%, what will be the new access time? • iii) What conclusions can be drawn about performance sensitivity?
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) What are the key characteristics of multiprocessor systems? Compare symmetric and asymmetric multiprocessing with block diagrams and real-world usage. (ii) What is inter-processor arbitration? Explain daisy-chaining and parallel arbitration methods used to resolve conflicts between processors. b) Differentiate between Programmed I/O, Interrupt-Driven I/O, and Direct Memory Access (DMA) in terms of speed, CPU involvement, and hardware complexity. Draw the data transfer process for each method. c) Describe the complete process of an interrupt cycle. Explain how multiple interrupt sources are handled using priority and vectored interrupt mechanisms.

EVEN SEMESTER EXAMINATION, 2024 – 25

2025 Year (II-Sem) B.C.A.

Environmental Studies

Duration: 3:00 hrs

Max Marks: 100

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20)
a)	(i) Explain the concept of ecological succession in an ecosystem. Give examples. (ii) How has man interfered with the ecosystem? Give an example of the same.
b)	What is an ecological pyramid? Explain about its types? Also write few limitations of the ecological pyramid.
c)	Describe how energy flows in an ecosystem. Compare and contrast food chains and food webs, and explain the concept of ecological pyramids with diagrams.
Q 2.	Answer any two parts of the following. (10x2= 20)
a)	(i) Differentiate between renewable and non-renewable natural resources. Provide two examples of each and briefly discuss their importance. (ii) List and explain any three major threats to biodiversity. How can community participation help in biodiversity conservation?
b)	Explain the various in-situ and ex-situ methods of biodiversity conservation. Assess the effectiveness of these approaches with suitable examples.
c)	Define biodiversity. Discuss its various values (ecological, cultural, economic) and give an overview of the global and Indian distribution of biodiversity.
Q 3.	Answer any two parts of the following. (10x2= 20)
a)	(i) What is incineration in terms of solid waste management? (ii) What is the greenhouse effect? How does it contribute to global warming?
b)	Define environmental pollution. Discuss the major types of pollution (air, water, soil, and noise) and explain modern strategies for effective waste management.
c)	What are radiation and nuclear hazards? Highlight their sources, environmental impacts, and safety measures to mitigate associated risks.
Q 4.	Answer any two parts of the following. (10x2= 20)
a)	(i) How can human society contribute to effective waste management? (ii) List any three important environmental laws in your country and their main purposes.
b)	Explain the significance of traditional wisdom and indigenous knowledge in ensuring livelihood security and sustainable natural resource management.
c)	Discuss key environmental legislations aimed at conservation and pollution control. How effective have these laws been in protecting the environment?
Q 5.	Answer any two parts of the following. (10x2= 20)
a)	(i) Name two major environmental movements and describe their main objectives. (ii) List and explain any two major human health risks associated with environmental pollution.
b)	Discuss the impact of rapid population growth on natural resources and the environment. Suggest measures to manage population sustainably.
c)	Critically examine the impact of human activities on the environment. Discuss how modern development practices can be aligned with environmental sustainability.

EVEN SEMESTER EXAMINATION, 2024 – 25

1st Year (IIInd Sem) B.C.A.

Object Oriented Programming using Java

Duration: 3:00 hrs

Max Marks: 100

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20)
a)	(i) How an array is declared in JAVA? Write a Java program that takes an array of 10 integers and prints the largest and smallest numbers. (ii) Discuss different types of loops in Java (for, while, do-while) with syntax and example programs.
b)	Differentiate between method overloading and overriding. Define a method named calculate Area to find the area of a rectangle and overload it to find the area of a square .
c)	Describe Java data types in detail. Write a program that declares all primitive data types and displays their values.
Q 2.	Answer any two parts of the following. (10x2= 20)
a)	(i) Explain the concept of a constructor in Java. How is it different from a method? (ii) Write a short note on visibility modifiers in Java (public, private , protected , default) with examples.
b)	What is the class and object? How we create class and object in JAVA ? Provide an example.
c)	What is string buffer? How does it differ from a string? Give the three ways of creating a string object.
Q 3.	Answer any two parts of the following. (10x2= 20)
a)	(i) Explain the use of the super keyword in Java with an example involving inheritance. (ii) What is the difference between an abstract class and an interface in Java?
b)	Create an abstract class Shape with an abstract method area() . Extend it with two classes Circle and Rectangle, and implement the area() method for both . Write a main method to demonstrate dynamic binding.
c)	Explain how packages are created and used in Java . Create a package named mypackage with a class Calculator, and access it from another class.
Q 4.	Answer any two parts of the following. (10x2= 20)
a)	(i) Describe the life cycle of an Applet with suitable method names. (ii) What are the differences between AWT and Swing in Java? Give examples.
b)	What is the role of layout managers in Java AWT ? Briefly explain Flow Layout, Border Layout, and Grid Layout.

	c) What is event delegation model in java? Write a program to create an Frame which contains three buttons having labels "one"," two" and "three" and one text field. When user click any button its label text appears in the text field.	
Q 5.	Answer any two parts of the following.	(10x2= 20)
	a) (i) Differentiate between text I/O and binary I/O in Java. (ii) List and explain the different states in the life cycle of a thread.	
	b) Write a Java program that handles Arithmetic Exception and Array Index Out Of Bounds Exception using try-catch-finally.	
	c) What is the difference between extending Thread and implementing Runnable interface? Demonstrate the creation of a new thread by extending the Thread class.	

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Sub Code: BCAT009

ROLL NO 241627000007

EVEN SEMESTER EXAMINATION, 2024 - 25

I Year (II Sem) B.C.A.

Software Engineering

Duration: 3:00 hrs

Max Marks: 100

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. a) (i) Describe the main components of software and their functions. (ii) What are software myths? Mention any two. b) Describe the Capability Maturity Model Integration (CMMI) and its levels. c) Compare and contrast the waterfall model and incremental process model.	(10x2= 20)
Q 2.	Answer any two parts of the following. a) (i) Differentiate between functional and non-functional requirements. (ii) What is the purpose of feasibility study in requirements engineering? b) Describe system models with emphasis on behavioral and data models. c) Describe the software requirements engineering process with a diagram.	(10x2= 20)
Q 3.	Answer any two parts of the following. (i) Explain the use of sequence diagrams in design. (ii) Define architectural style and give examples. b) Describe the role of UML in design. Explain different UML diagrams used in software architecture. c) Explain the design process and design quality attributes in software engineering.	(10x2= 20)
Q 4.	Answer any two parts of the following. a) (i) What is debugging? Mention any two debugging techniques. (ii) Define validation and system testing. b) Differentiate between black-box and white-box testing with examples. c) Explain how software metrics help in maintaining and improving product quality.	(10x2= 20)
Q 5.	Answer any two parts of the following. a) (i) What is the RMMM plan? (ii) Define software risk. List any two types. b) Discuss the ISO 9000 quality standards and their relevance to software engineering. c) What is statistical software quality assurance? How does it differ from formal reviews?	(10x2= 20)

Sub Code: BCAT006

ROLL NO. 241627000007

EVEN SEMESTER EXAMINATION, 2024 – 25

I Year (II Sem) B.C.A.

Data Structures

Duration: 3:00 hrs

Max Marks: 100

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Define data structure and explain its types. (ii) Explain the term "sub-algorithm" with an example. b) Discuss the different asymptotic notations used to describe the complexity of algorithms. c) Explain the concept of time and space complexity.
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) What is a Stack Data Structures? Also explain its operations. (ii) Differentiate between singly linked lists and doubly linked lists with examples. b) What are circular queues? Explain their array representation. c) For an array A = [2, 4, 6, 8, 10], calculate the address of the 4th element if the base address is 1000 and each element takes 4 bytes of memory.
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Explain Tree and its terminologies. (ii) Differentiate between Binary tree and Binary Search Tree b) Explain the concept of AVL trees and their significance in maintaining balanced trees. c) Given a binary search tree, insert the elements 15, 10, 20, 8, 12, 17, 25 into it and draw the resulting tree.
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) Define a graph and explain the difference between directed and undirected graphs. (ii) What is a spanning tree? How is it different from a graph? b) Describe the adjacency matrix representation of graphs with an example. c) Explain Depth-First Search (DFS) and Breadth-First Search (BFS) techniques.
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) Write the algorithm for the bubble sort technique. (ii) What is hashing? Discuss the different types of hashing schemes. b) What is Sorting and searching? c) Compare and contrast merge sort and quick sort techniques.
